GPunch

Secure On-Site Presence Verification System

Kavin M · Shanmugappriya K

PSG College of Technology · MCA · Dr. N. Ilayaraja

May 2026

Agenda

#	Section	Time
1	Introduction — Scope, Value, Audience, Use, Architecture	~3 min
2	Functional Requirements — 13 FRs + Test Cases	~4 min
3	External Interface Requirements	~2 min
4	Non-Functional Requirements	~3 min
5	Definitions & Acronyms	~1 min
	Q & A	~2 min

1 · Introduction — Problem & Scope

Why existing systems fail

System	Weakness
Paper registers	Easily forged; no audit trail
PIN / password	Credentials shared — buddy punching
Selfie check-in	Privacy concerns; photo-spoofable
Client-side GPS	Overridden by mock-location apps

GPunch scope

- **Backend** REST API — Node.js 18 + Express + MongoDB Atlas
- **Android Client** — Kotlin (API 26+), GPS + OTP + device binding
- Single campus, single geofence, fully passwordless

1.2 · Product Value — Three Security Pillars

```
+--------------------+  +--------------------+  +--------------------+
|  WHO are you?      |  |  WHICH device?     |  |  WHERE are you?    |
|                    |  |                    |  |                    |
|  Institutional     |  |  ANDROID_ID        |  |  GPS Geofence      |
|  Email OTP         |  |  Device Binding    |  |  Haversine (srv)   |
|                    |  |                    |  |                    |
|  No password       |  |  One device only   |  |  Campus boundary   |
+--------------------+  +--------------------+  +--------------------+
```

- **Zero buddy-punching** — each account locked to one physical device
- **Zero GPS spoofing impact** — server re-validates coordinates
- **Zero attendance data loss** — WorkManager offline queue

1.3 & 1.4 · Intended Audience & Use

Audience

Role	Purpose
Employee	Register → OTP → Clock In/Out via Android app
Admin	Configure geofence, manage devices, view audit logs, export CSV
System	Auto clock-out, offline retry (background automation)

Typical employee flow (30 seconds)

1. Open app → tap **Clock In**
2. App checks GPS is on → obtains location fix
3. Server validates Haversine distance against geofence
4. Success → foreground service monitors in background

```
+---------------------+   HTTPS/REST   +---------------------+
|  Android Client     | <------------> |  Express REST API   |
|  (Kotlin, API 26+)  |                |  (Node.js 18)       |
|                     |                +--------+------------+
|  Activities (MVVM)  |                         |
|  GeofenceService    |                +--------v------------+
|  PendingClockOut    |                |   MongoDB Atlas     |
|  WorkManager        |                |  Users · Attend.    |
|  SessionManager     |                |  Geofence · Audit   |
+---------------------+                +---------------------+
                                                |
                                        Gmail SMTP (OTP)
```

Backend: Route Handlers → Middleware (JWT/rate-limit) → Mongoose Models → Atlas

Android: View (Activity) → ViewModel (LiveData) → Repository (Retrofit) → Service/Worker

2 · Functional Requirements — Authentication

FR	Feature	Key Rule
FR-01	Domain-restricted registration	Only `@psgtech.ac.in` emails accepted; rejects log `INVALID_DOMAIN`
FR-02	OTP verification + device binding	Strict string equality; binds `ANDROID_ID` on success; issues JWT
FR-03	Device binding enforcement	Login from different device $\rightarrow$ HTTP 403 + `UNAUTHORIZED_DEVICE` log

Auth flow

```
Register (email)  -->  OTP email  -->  Enter OTP  -->  Device bound  -->  JWT
Login (email)     -->  Device check  -->  OTP email  -->  Enter OTP  -->  JWT
```

Passwordless throughout — no password field exists anywhere in the system

FR	Feature	Key Rule
FR-04	GPS settings check	`LocationSettingsRequest` → system dialog if GPS off
FR-05	Geofence clock-in	Server Haversine; inside → HTTP 201; outside → HTTP 403 + distance
FR-06	Mock GPS detection	`location.isMock` → deny + log `MOCK_LOCATION`
FR-07	Background monitoring	Foreground service; adaptive poll 15 s (moving) / 60 s (still)
FR-08	Auto clock-out	Service detects boundary exit → API call → push notification
FR-12	Offline queue	Network failure →

2 · Functional Requirements — Admin & Test Cases

Admin capabilities (FR-09 to FR-13)

- **FR-09** Set geofence lat/lng/radius/domain (changes are instant)
- **FR-10** Reset device binding — unlinks lost/replaced phone
- **FR-11** View paginated audit logs (filterable by event type)
- **FR-13** Export full attendance CSV

Test case summary (14 / 14 passing)

TC	Covers	Result
TC-01 & TC-02	Domain reject / accept	Pass
TC-03 & TC-04	OTP failure / device mismatch	Pass
TC-05 – TC-10	GPS check, geofence, mock, background, auto-out	Pass
TC-11 – TC-14	Admin config, device reset	Pass

3 · External Interface Requirements

3.1 User Interface (Android)

Screen	Purpose
Login / Register	Passwordless entry; domain validated on register
OTP Verification	6-digit input; 60 s resend countdown
Dashboard	Clock In/Out, distance indicator, session timer, history

3.2 Hardware · 3.3 Software · 3.4 Communication

Layer	Technology
GPS	Fused Location Provider (Google Play Services 21.3.0)
Background jobs	AndroidX WorkManager 2.9.0
HTTP client	Retrofit 2.11 + OkHttp 4.12 (JWT interceptor)
Database	MongoDB Atlas via Mongoose 8.3

4 · Non-Functional Requirements — Security & Capacity

4.1 Security highlights

Threat	Mitigation
Buddy punching	ANDROID_ID device binding — one account, one device
GPS spoofing	`location.isMock` (client) + Haversine server re-validation
NoSQL injection	Explicit typed Mongoose queries + `eventType` whitelist
OTP brute-force	20 req / 15 min rate limit; 10-minute OTP expiry
JWT forgery	HS256 + long secret; 7-day expiry

4.2 Capacity · 4.3 Compatibility

- **500+ users** on MongoDB Atlas M0 free tier (single campus)

4 · Non-Functional Requirements — Reliability to Usability

4.4 Reliability

- WorkManager payload survives **app restart + device reboot** (FR-12)
- `BootReceiver` restores foreground service after reboot if clocked in

4.5 Scalability

- Horizontal scaling: stateless Node.js + Atlas connection pooling
- Rate-limit config externalised to `.env` — no code change to scale

4.6 Maintainability

- Backend: **MVC** — `models/ routes/ middleware/ utils/`
- Android: **MVVM** — Activity → ViewModel → Repository → Service

4.7 Usability · 4.8 Other

- **No password** — name + email + 6-digit OTP only
- Human-readable errors; one-tap notification → Dashboard

5 · Definitions & Acronyms

Term	Meaning
OTP	One-Time Password — 6-digit code, 10 min expiry, email-delivered
JWT	JSON Web Token — HS256 signed, 7-day session token
Geofence	Virtual boundary: centre (lat/lng) + radius in metres
Haversine	Great-circle distance formula — server-side, client cannot override
ANDROID_ID	Device fingerprint bound to account on first OTP verify
FR / NFR / TC	Functional Req. / Non-Functional Req. / Test Case
FLP	Fused Location Provider — Google Play Services GPS API

Summary

GPunch delivers three guarantees

Who *you are (OTP)* · ***Which*** *device (ANDROID_ID)* · ***Where*** *you are (Haversine)*

Component	Technology	Status
Backend API	Node.js · Express · MongoDB Atlas	10 tests passing
Android App	Kotlin · MVVM · WorkManager	14 TC verified
Security	JWT · helmet · rate-limit · audit logs	All threats mitigated
Documentation	SRS (IEEE + 23MX21) + Presentation	Complete

All 14 test cases pass

FR-01 through FR-13 · TC-01 through TC-14 · Zero known vulnerabilities

Thank You

GPunch — Secure On-Site Presence Verification

Kavin M · Shanmugappriya K

MCA · PSG College of Technology

Faculty Guide: Dr. N. Ilayaraja

Questions?